Requirement Document: N8N Hands-On Beginner Course

Topic 7: Schedule Trigger

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 7 - Schedule Trigger
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
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2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees and employees how to use the Schedule Trigger node to run workflows automatically using time-based intervals and custom cron expressions. Trainees will learn the basics of cron syntax and configure multiple schedule rules within a single trigger node.

2.2 Learning Scope

In Scope:

- Adding and configuring the Schedule Trigger node within a workflow.
- Setting up simple daily intervals at a specific time (e.g., 09:00 AM).
- Writing and explaining custom cron expressions (e.g., triggering weekdays at 8:30 AM).
- Understanding the standard 5-field cron syntax format (minute hour day-of-month month day-of-week).

- Connecting the trigger to a downstream action node (Edit Fields) to capture and log execution timestamps.

Out of Scope:

- Configuring timezone adjustments dynamically per-execution (using external timezone APIs).
- Triggering external events via Webhooks or Polling nodes (covered in separate modules).
- Configuring complex error-handling or retry policies on schedule failures.

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 through Topic 6.
- A basic understanding of time representation (hours, minutes, days of the week).
- An active n8n instance with workspace edit and testing permissions.

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Simple Daily Interval	The workflow shall trigger automatically on a daily basis at a specific hour and minute (e.g., daily at 09:00 AM).	**High**
REQ-002	Custom Cron Schedule	The workflow shall trigger automatically on a custom schedule representing weekdays (Monday through Friday) at a specific time (e.g., 08:30 AM).	**High**
REQ-003	Cron Syntax Mapping	The custom schedule must use a standard 5-field cron expression equivalent to `30 8 * * 1-5` (representing minute 30, hour 8, every day of month, every month, days 1 to 5 / Mon-Fri).	**High**
REQ-004	Downstream Logger	The Schedule Trigger node shall connect to an Edit Fields (Set) node named Log Execution to record each automatic execution.	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-005	Execution Timestamp	The Log Execution node shall output a single dynamic string property `executionTime` containing the timestamp of when the trigger fired using n8n expression syntax referencing the current date and time.	**High**
REQ-006	Workflow Activation	The finished workflow must be capable of being toggled to Active to run automatically in the background according to the defined schedules.	**Medium**

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Cron Execution Engine	The underlying n8n instance scheduler must parse the standard 5-field cron expression syntax in real-time.
TR-002	Multiple Schedule Rules	The Schedule Trigger node must support configuring multiple schedule rules (e.g., a simple daily rule and a custom cron rule) within the same node's configuration parameters.
TR-003	Dynamic Variable Reference	The workflow expressions engine must resolve the dynamic `$now` or `$dateTime` helper variables to format the output execution timestamp.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to understand how to schedule workflows automatically to replace manual routines, a critical skill for business process automation.

Persona	Role	Need
Instructor / Reviewer	Evaluation	Needs to verify that the student understands basic cron fields and has configured both simple and custom cron triggers in the Schedule Trigger node.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The n8n instance's system timezone is configured correctly, ensuring the time-based triggers fire at the trainee's expected local time.
- **Assumption 2 (ASM-002):** The workflow is saved and active to ensure background schedule triggers are evaluated by n8n.
- **Constraint 1 (CON-001):** The execution resolution is subject to the cron engine's precision, which operates on a minimum granularity of minutes.